

Exploring the nexus between CEO characteristics and the value of excess cash holdings through the lens of the resource-based view theory

Moncef Guizani and Chouayb Larabi

Moncef Guizani and Chouayb Larabi are both based at the Faculty of Business Studies, Arab Open University, Riyadh, Saudi Arabia.

Abstract

Purpose – This study aims to examine the relationship between CEO characteristics and the value of excess cash holdings from the perspective of resource-based view (RBV) theory in the context of Malaysia.

Design/methodology/approach – The analyses were made using ordinary least squares across 173 non-financial firms listed in Bursa Malaysia over the period of 2015–2021. The authors address potential endogeneity through the generalized method of moments. The results are also robust to alternative measures of excess cash holdings.

Findings – The results showed that female CEOs and CEOs' educational level are significantly positively related to the value of excess cash holdings. In contrast, CEO tenure and CEO age negatively affect a firm's excess cash valuation. The results are robust to measurement error and endogeneity issues.

Practical implications – The empirical results have useful policy implications. For practitioners, firms are recommended to prioritize the selection of female CEOs and CEOs with high education levels within their top management, as this initiative can result in improved value associated with excess cash holdings. In addition, policymakers are recommended to guide programs that attempt to improve educational attainment and gender diversity in business leadership. This study also provides investors with insightful information about the possible relationship between CEO traits and company performance, especially with regard to measures for managing surplus capital.

Originality/value – To the best of the authors' knowledge, this study is the first to explore the role of CEO characteristics in the value of excess cash holdings based on the RBV theory.

Keywords CEO characteristics, Excess cash holdings, Firm value, Resource-based view theory

Paper type Research paper

1. Introduction

Corporate cash holdings are a crucial factor that can significantly affect the performance of firms. This impact, however, is not always positive (Belkhir *et al.*, 2014; Jilani *et al.*, 2023). On the one hand, in the presence of financial constraints, firms are inclined to hold a threshold of cash to face high external financing. Within this context, cash holdings are considered to be valuable and can have a positive impact on firm value (Harford *et al.*, 2008; Amess *et al.*, 2015). However, holding excess cash might be at the expense of shareholders' interests, leading to agency costs resulting from shareholder–manager conflicts (Jensen, 1986). If excess cash is not used efficiently, it will be a strong candidate for being turned into managers' private benefits (Dittmar and Mahrt-Smith, 2007; Seifert and Gonenc, 2018). Hence, managers need to consistently decide whether to disgorge cash to shareholders, spend it internally, expend it on external acquisitions or continue to hold it (Harford *et al.*, 2008; Jabbouri and Almustafa, 2021).

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Earlier studies have widely discussed the value relevance of cash holdings. They mainly focused on examining cash holding policy aspects, highlighting the role of corporate governance in determining the use and eventual valuation of cash holdings. For instance, [Dittmar and Mahrt-Smith \(2007\)](#) find that firms with high excess cash and poor governance experience deficient operating performance. Likewise, [Harford et al. \(2008\)](#) report that cash is less valuable when managerial agency problems are severe. They find that firms with weaker corporate governance structures hold more cash but spend cash quickly on investment rather than grow cash reserves. On the contrary, some evidence supports a more positive role for cash holdings in value creation. For example, [Jabbouri and Almustafa \(2021\)](#) argue that the positive impact that national governance has on individual firms by reinforcing investors' protection and lowering agency problems increases the added value of cash holdings. Similarly, [Ilyas et al. \(2023\)](#) show that strong corporate governance positively affects the contribution of excess cash holdings to firm value. As such, the amount of cash held provides an indirect indication of the corporate governance strength of a company.

The main argument of this research is that CEOs' personal characteristics play a vital role in corporate governance and financial decisions (e.g. [Ben Mohamed et al., 2014](#); [Xie, 2015](#); [Gupta et al., 2018](#); [Rahman and Chen, 2023](#)). Therefore, the relationship between corporate cash holdings and firm value may be contingent on the social-behavioral attributes of the CEO.

Recently, a large amount of cash was reported in the financial statements of companies worldwide ([Dao and Maggi, 2018](#)). Accordingly, it appears to be an imperative issue to identify the characteristics of managers for optimal use of resources and better results in terms of a firm's performance ([Gupta et al., 2021](#)). Prior research posited that the psychological and social characteristics of CEOs can be approximated by demographics such as tenure, education, age and gender ([Hambrick, 2007](#)).

Previous studies used different theoretical perspectives, including the trade-off theory ([Opler et al., 1999](#)), the pecking order theory ([Myers and Majluf, 1984](#)) and the free cash flow (agency) theory ([Jensen, 1986](#)), to investigate the corporate cash holding decision. However, relatively scant attention is paid to the nexus between CEO characteristics and cash-holding valuation. Accordingly, to address this gap and contribute to the extant literature, the current study aims to examine whether and how CEO characteristics have bearings on excess cash value from the perspective of the resource-based view (RBV) theory.

Existing research within the RBV literature reports that the value creation of companies is influenced by nonfinancial resources (NFRs) ([Hristov et al., 2024](#)). NFRs, such as technological, knowledge and human resources, play a fundamental role in achieving sustainable competitive advantage ([Basri and Arafah, 2020](#)). RBV theory emphasizes that firm resources should be valuable, rare, inimitable and non-substitutable ([Barney, 1991](#)). From this perspective, the RBV can be applied to analyze how specific CEO characteristics are considered valuable resources contributing to the firm's financial decisions. CEO characteristics, such as experience, expertise, tenure, decision-making skills and risk preferences, can be viewed as valuable resources. If a CEO possesses characteristics that contribute to effective financial management, such as optimizing cash holdings for strategic purposes, it can lead to a competitive advantage. This theory suggests that firms with CEOs possessing advantageous characteristics in financial decision-making may achieve better financial performance than those without such resources.

We chose Malaysia as a research context for three reasons. First, firms in developing markets hoard substantial cash reserves, and the chances of private benefits are higher, more accessible and also typically riskier in these markets. Second, as [Akhtar \(2023\)](#) mentioned, Malaysian companies maintain more cash, amounting to 12.19% of total assets

on average, than other growing financial markets such as Indonesia, Thailand and the Philippines. Third, in Malaysia, the principles and best practices of good governance are described in the Code on Corporate Governance (MCCG). The MCCG is an important mechanism for corporate governance reform, adopting international standards and positively influencing firms' practices. The MCCG urges the separation of the duties of Chairman and CEO to promote good governance among the listed firms in Malaysia. To achieve the operating performance goal, the CEO is urged to provide support for the Chairman in discharging his or her duties in managing the operating performance (Chandren *et al.*, 2021).

The study contributes to the literature in several ways. First, this paper contributes to the extensive debate on corporate liquidity by documenting the role of personal characteristics of CEOs, such as gender, tenure, age and education, on the valuation of excess cash holdings. While earlier studies have extensively discussed the role of corporate governance in the valuation of cash holdings, this study focuses on CEOs' social-behavioral attributes as essential factors influencing the value of excess cash holdings. Second, the present research attempts to explain the nexus between CEO characteristics and the value of excess cash holdings by featuring the RBV theory. The key feature of our model is that CEO traits can be viewed as valuable resources that will enhance corporate valuation and performance. Third, this research contributes to the corporate governance literature because it considers a developing country (Malaysia), which has different economic, legal, social and cultural environments than developed countries, especially the Western countries where many previous studies have been conducted. Therefore, we argue that the findings from Malaysia will be beneficial to other countries that have similar institutional settings. Finally, our findings support shareholders, debt holders and investors in controlling companies' excessive cash holdings.

The rest of this article is organized as follows. The next section provides theoretical background and hypotheses. In Section 3, we develop our hypotheses. The research methodology, encompassing aspects such as data collection, sample selection, variable definitions and specifications of regression models, is exhaustively elucidated in Section 4. Section 5 reports and provides a discussion on the outcomes of the regressions. Section 6 presents the conclusion, elaborates on the implications and limitations of the study and provides suggestions for the venue of future research.

2. Theoretical framework

The impact of CEO characteristics on the value of excess cash holdings is explained from the theoretical perspective of RBV theory. Previous studies used different theoretical perspectives, including agency theory (Pinkowitz *et al.*, 2006; Dittmar and Mahrt-Smith, 2007; Frésard and Salva, 2010; Belkhir *et al.*, 2014), pecking order theory (Elbadry *et al.*, 2015) and upper echelon theory (Han *et al.*, 2017), to investigate the relationship between corporate governance characteristics and the value of cash. Recently, strategic management researchers focused on why some firms are more profitable than others (Barney and Clark, 2007). One approach is the resource school that seeks competitive advantages inside the organization, particularly internal resources (Galbreath, 2005). The RBV attributes relevance to internal aspects, and the central premise is that each organization's performance is based on its unique resources. Barney (1991) and Wernerfelt (1984) indicated that resources are internal attributes, including financial assets, capabilities, processes, routines and knowledge, tied semi-permanently to or controlled by a firm. Resources and capabilities are valuable as they improve the organization's efficiency and effectiveness. In addition, reviews have pointed to human factors as a source for creating and maintaining competitive advantages (Hoskisson *et al.*, 1999).

Within the RBV framework, CEO characteristics are often categorized as part of human capital. Therefore, the characteristics of CEOs can help the firm obtain tangible resources,

such as cash holdings. Firms maintain cash holdings as a form of liquidity to meet short-term obligations and unforeseen expenses. Adequate liquidity can enhance a firm's ability to navigate financial challenges, contribute to financial stability and reduce the risk of financial distress.

Furthermore, firm performance is sequentially influenced by strategic choices triggered by managerial background characteristics. Therefore, influenced by their characteristics, CEOs may choose strategic actions within their available free cash flow, such as acquisitions, divestitures, financial choices and financial resource allocation, which, in turn, influence firm value.

3. Literature review and hypotheses development

Excessive cash hoarding by firms is a serious concern for shareholders. Although firms need adequate amounts of cash for their operational, financing and investment needs, accumulating cash above the optimal level may increase the risk associated with its expropriation (Jensen, 1986; Harford *et al.*, 2008; Ilyas *et al.*, 2023). Existing literature provides ample evidence indicating that the contribution of cash holdings to firm value primarily relies on their deployment by corporate managers. When corporate governance is better and the interests of shareholders and management are more aligned, any cash withheld should be more likely to be spent on positive net present value (NPV) projects, such that cash is more valuable under such circumstances (Dittmar and Mahrt-Smith, 2007; Harford *et al.*, 2008).

Research has shown that the CEO's characteristics can have a significant impact on corporate financial policy, especially when it comes to cash policy. For instance, Han *et al.* (2017) examine the effect of CEO characteristics on the value of excess cash using the listed firm data in Korea. They find that the value of excess cash depends on CEO characteristics such as educational background and CEO demographics. Using US non-financial firms as the study sample, Belkhir *et al.* (2014) provide evidence that excess cash holdings contribute less to firm value when shareholders expect their value to be destroyed because of managers' conservative behavior. Even though these studies focused on the relationship between CEO characteristics and the value of cash holdings from the perspective of corporate governance, this research focuses on the theoretical framework of RBV theory, which states that CEO characteristics that are valuable, rare, or non-substitutable by other firms could improve the valuation of excess cash holdings.

3.1 CEO gender

The RBV theory recognizes that the presence of female directors in top management helps firms capitalize on their access to vital resources through their skills, competencies, knowledge and leadership styles, which are different from those of male directors (Hillman *et al.*, 2007). According to Charles *et al.* (2018), women are viewed as valuable human resources and contribute diverse perspectives, aligning with the theory's application to comprehending gender diversity in the business area. The RBV theory claims that firms with female CEOs mainly accrue benefits via three channels: advisory, legitimacy and access to critical resources and communication channels (Liu *et al.*, 2014). For instance, firms with female CEOs are likely to sustain good relationships with female clients or consumers, thereby significantly contributing to firms' performance (Saggar and Singh, 2017).

Prior empirical literature offers mixed results on the impact of female CEOs on cash holdings. For instance, La Rocca *et al.* (2019) report that female executives are more likely to hold more cash compared to their male counterparts because of the precautionary motive of unexpected financing. Suherman *et al.* (2021b) suggest that female executives are more conservative and risk-averse, thereby holding more cash with a precautionary

motive. Consequently, the firm might lose investment opportunities, resulting in decreasing shareholders' value. On the contrary, some evidence supports a positive role for female CEOs in the valuation of excess cash holdings. For example, [Gul et al. \(2011\)](#) indicate that female leaders improve investment decisions, provide more transparent financial information and increase shareholder value. Likewise, [Levi et al. \(2014\)](#) find that female directors alleviate the problems of suboptimal investment. Thus, the chances of wasting excess cash flow will be mitigated. [Ullah et al. \(2020\)](#) find that female CEOs are associated with efficient investment decisions. [Guizani and Abdalkrim \(2022\)](#) further argue that female CEOs are more likely to reduce the potential expropriation of firm resources. Additionally, [Amore et al. \(2014\)](#) show that cooperation on the boards between female CEOs and female directors may result in more efficient managerial decision-making. Therefore, female CEOs significantly reduce the misallocation of corporate resources by controlling corporate insiders.

In Malaysia, the MCCG requires 30% female directors for public businesses. In total, 30% female board presence will increase the variety of talents, experiences, viewpoints and tactics, improving board members' effectiveness and corporate governance ([Chandren et al., 2021](#); [Guizani and Abdalkrim, 2022](#)). Using a sample of Malaysian listed firms, it was found that female directors improve resource allocation by enhancing governance practices.

From the above discussion, the following hypothesis is proposed:

H1. There is a positive relationship between female CEOs and the value of excess cash.

3.2 CEO tenure

In line with RBV theory, highly skilled CEOs are expected to unambiguously maximize the firm's output by better allocating the firm's valuable resources, as they are one of the human strategic resources associated with positive financial outcomes ([Hambrick, 2007](#)). CEOs with extensive experience may be relatively rare. CEOs with extended tenure may deeply understand the organization's financial history, cash flow patterns and risk profile, which can influence cash management decisions.

Prior studies presented mixed findings. For instance, [Lim and Lee \(2019\)](#) show that as a CEO's tenure increases, their experience, expertise and knowledge also expand, leading to better financial resource allocation and firm performance. Therefore, prolonging a CEO's appointment can be a motivating factor to initiate efficient strategies to improve business performance and uphold their reputation ([Suherman et al., 2021b](#)). Furthermore, long-tenured CEOs are more likely to invest their financial resources into projects that may potentially generate positive returns ([Lim and Lee, 2019](#)). By contrast, other studies address the self-interest of the long-tenured CEO to explain the deteriorating performance of excess cash holdings. In this regard, long-tenured CEOs in firms with excess cash reserves can waste excess cash, lowering the marginal value of cash. As argued by [Ben Mohamed et al. \(2014\)](#), when CEOs have a long tenure, they can exploit internal funds to finance their preferred investment projects. This could lead to excessively using internal cash flow to invest even in projects with a negative NPV. [Han et al. \(2017\)](#) further provide evidence that as the tenure of CEOs lasts longer, the value of excess cash decreases. Like other emerging market economies, Malaysian companies are dominated by controlling shareholders in a concentrated ownership structure ([AlQadasi and Abidin, 2018](#)). As argued by [Claessens et al. \(2000\)](#), companies with concentrated ownership are run by the major shareholders, who play a significant role in their decision-making processes. Consequently, the firm's CEO is appointed for a long period, thus leading to accumulated experience, expertise and knowledge. In this regard, [Alzeban \(2022\)](#) reports that Malaysian firms with CEO experience exhibit a higher level of implementation of internal audit

recommendations and that management support for internal audit is positively associated with corporate performance. Thus, we hypothesize:

H2. There is a positive relationship between CEO tenure and the value of excess cash.

3.3 CEO age

Based on RBV theory, CEO age is a necessary “firm’s human resource” that improves competitive advantages. The CEO’s accumulated experience and knowledge developed over years that typically come with age can be considered valuable resources. Older CEOs have a greater competitive advantage and working experience than younger CEOs (Chandren *et al.*, 2021). The expertise gained over years of professional experience becomes a resource that contributes to the CEO’s ability to navigate financial challenges and optimize cash management strategies.

Previous literature provided two opposing perspectives on the effect of CEO age on firm value. On the one hand, Hambrick (2007) suggests that older CEOs tend to be more conservative and may not be as willing to propose new ideas as they are comfortable with their current leadership style and how they run their companies. Bertrand and Schoar (2003) find that CEOs from older generations appear less aggressive, opting for lower capital expenditures and higher cash holdings, which ultimately reduce firm value. On the other hand, Serfling (2014) reports that older CEOs reduce firm risk through less risky investment policies. Wang *et al.* (2016) argue that CEOs may have stronger linkages and better access to resources, resulting in better firm performance. Gupta (2022) finds that older CEOs reduce the sensitivity of investment to internal cash flow. This suggests that older CEOs reduce financial constraints and, therefore, the cost of external financing. In Malaysia, Razali *et al.* (2022) find that CEO age in the technology industry has a positive relationship with firm value. Further, Ahmad *et al.* (2022) report that CEO age and firm performance are positively correlated. The positive relationship between CEO age and corporate performance confirms that older directors are effective leaders in managing a firm’s financial resources. Thus, the following hypothesis is proposed:

H3. There is a positive relationship between CEO age and the value of excess cash.

3.4 CEO educational level

According to RBV theory, educational level is crucial to enhancing competitive advantages and the overall performance of firms. Barney (1991) suggests that having an educational background can help companies make informed strategic decisions and achieve greater competitive advantages. King *et al.* (2016) report that educational background affects the development of a CEO’s personal and cognitive values, which help increase the decision-making process’s usefulness. A CEO with the right education, knowledge and talents could be a source of competitive advantage and improve the firm’s performance (Orazalin *et al.*, 2016). According to Naseem *et al.* (2020), education helps CEOs make effective choices that can enhance firm value. Gupta *et al.* (2021) document that a CEO’s education plays a significant role in corporate investment decisions. Their findings reveal that the CEO’s financial education reduces the financial constraints faced by the firm, thereby declining ICFS and having a positive impact on corporate investment. Regarding the value of cash holdings, Han *et al.* (2017) find that an MBA CEO and a master CEO increase the value of excess cash. In Malaysia, several studies confirm the role of educational level in boosting operating performance (Ahmad *et al.*, 2022; Hasnan *et al.*, 2023). Furthermore, Chua *et al.* (2020) report that when firms are managed by highly educated CEOs, agency conflict between shareholders and managers reduces; hence, managers’ capital structure goals will be aligned with the shareholders at optimal leverage.

From the above discussion, the following hypothesis is stated:

- H4.* There is a positive relationship between CEO educational level and the value of excess cash.

4. Data and methodology

4.1 Data sources and sample of the study

Data were collected from the firms listed under the Capital Market Development Fund – Bursa Research Scheme (CBRS). CBRS is a joint effort between the Capital Market Development Fund (CMDf) and the Bursa research scheme “to raise the profile of small and medium firms listed on Bursa Malaysia.” The research variables’ data were collected from the annual reports of 285 firms spanning seven years from 2015 to 2021. The reason for selecting a database of 285 companies was the availability of data covering all factors examined in the study. We have excluded the financial and banking companies from the analysis because their financial data is different in reporting because of business and regulatory factors. We have also excluded observations with negative excess cash holdings. The companies for which the data were unavailable for all seven years have been dropped. All continuous variables were winsorized at the 1st and 99th percentiles to mitigate the influence of outliers. Our final sample consists of 1098 firm-year observations and represents 173 firms. It covers six main economic sectors: construction, consumer products, industrial products, plantations, properties and technology. Accounting and stock market data were obtained from secondary sources, including yearly reports, the Bursa Malaysia website and DataStream. [Table 1](#) presents the sample selection and its distribution among industries.

4.2 Variables and measurement

4.2.1 Excess cash holdings. We follow previous studies (e.g. [Dittmar and Mahrt-Smith, 2007](#); [Frésard and Salva, 2010](#); [Ilyas et al., 2023](#)) to define excess cash holdings (ExCash) as the amount of cash holdings above a predicted optimal level. Our analysis focuses on firms with positive excess cash holdings that are not required for their day-to-day operations and investments.

In line with [Dittmar and Mahrt-Smith \(2007\)](#) and [Ilyas et al. \(2023\)](#), the following model is estimated to predict the normal cash holding level:

Table 1 Sample selection and distribution		
		Total firms
<i>Panel A: Sample selection</i>		
Total firms available to be sampled		285
Less:		
Financial companies		12
Firms with negative excess cash holdings		40
Firms with insufficient financial data		60
Final selected sample		173
<i>Panel B: Industrial composition of the sample</i>		
Industry classification	No. of firms	Percentage of sample
Construction	9	5.20
Consumer products	46	26.59
Industrial products	67	38.73
Plantations	10	5.78
Properties	28	16.18
Technology	13	7.52
Total	173	100
Source: Authors' own creation		

$$\begin{aligned} \text{Ln}(\text{Cash}_{i,t}) = & \beta_0 + \beta_1 \widehat{\text{MTB}}_{i,t} + \beta_2 \text{Ln}(\text{TA}_{i,t}) + \beta_3 \text{CFLOW}_{i,t} + \beta_4 \text{NWC}_{i,t} + \beta_5 \text{CAPEX}_{i,t} \\ & + \beta_6 \text{LEV}_{i,t} + \beta_7 \text{RD}_{i,t} + \beta_8 \text{DIV}_{i,t} + \beta_9 \text{Ln}(\text{Firm_Age}_{i,t}) + \text{YearFE} + \text{FirmFE} + \varepsilon_{i,t} \end{aligned} \quad (1)$$

We compute cash holdings as the natural logarithm of the ratio of cash and marketable securities to total assets ($\text{Ln}(\text{Cash})$). $\widehat{\text{MTB}}$ is the predicted market-to-book ratio (MTB), which is derived based on the average of lagged sales growth variables. MTB is instrumented to mitigate the potential endogeneity problem in the above model. TA is total assets, CFLOW is operating income minus interests and taxes, NWC is net working capital, which equals working capital minus cash, CAPEX is the ratio of capital expenditure, LEV is the ratio of total debt to total assets, RD is the research and development expenses to total assets and DIV is the common dividend paid divided by total assets. Firm_Age is firm age in years. To control for periodic and firm-specific omitted factors, [equation \(1\)](#) also includes year-fixed effects denoted by YearFE and firm-fixed effects represented by FirmFE.

Residuals from the above model are generated. Excess cash (ExCash) is derived by calculating the natural exponential of the residual, which is the percentage of deviation from the predicted cash ($\text{Ln}(\widehat{\text{CASH}})$).

4.2.2 CEO characteristics. This research used four dimensions of CEO characteristics, including CEO gender, CEO tenure, CEO age and CEO educational level. To measure CEO gender, a dummy variable, CEO_Fem, was used, which equals 1 if the CEO is a woman and 0 if the CEO is a man. CEO tenure is calculated as the number of years that have passed since a CEO was appointed up until the current fiscal year ([Lim and Lee, 2019](#); [Suherman et al., 2021b](#)). We include CEO tenure (CEO_Tn) in our analysis as a 1–0 dummy by splitting the sample into terciles: the highest CEO tenure tercile is coded as 1, the lowest tercile is coded as 0 and the middle tercile is discarded. CEO age is a ratio scale that indicates the CEO's age at the time of our observation period; age is measured in years ([Naseem et al., 2020](#); [Suherman et al., 2021b](#)). We include CEO age (CEO_Age) in our analysis as a 1–0 dummy by splitting the sample into terciles: the highest CEO age tercile is coded as 1, the lowest tercile is coded as 0 and the middle tercile is discarded. We use a dummy variable to measure CEO education (CEO_Edu), which takes the value 1 for postgraduates (master's and doctorate) and 0 for undergraduates and others ([Suherman et al., 2021b](#)).

4.3 Research design

Because the data in this study comprises multiple firms and spans several years, the panel data model is used to test the hypotheses. Following the work of prior seminal liquidity studies ([Pinkowitz et al., 2006](#); [Frésard and Salva, 2010](#); [Ilyas et al., 2023](#)), different specifications of the next panel data model were estimated:

$$\begin{aligned} \text{MV}_{i,t} = & \beta_0 + \beta_1 \text{CEO_Char}_{i,t} + \beta_2 \text{CEO_Char}_{i,t} \times \text{ExCash}_{i,t} + \beta_3 \text{ExCash}_{i,t} + \beta_4 \text{EAR}_{i,t} \\ & + \beta_5 \Delta \text{EAR}_{i,t} + \beta_6 \Delta \text{EAR}_{i,t+1} + \beta_7 \text{RD}_{i,t} + \beta_8 \Delta \text{RD}_{i,t} + \beta_9 \Delta \text{RD}_{i,t+1} + \beta_{10} \text{DIV}_{i,t} \\ & + \beta_{11} \Delta \text{DIV}_{i,t} + \beta_{12} \Delta \text{DIV}_{i,t+1} + \beta_{13} \text{INT}_{i,t} + \beta_{14} \Delta \text{INT}_{i,t} + \beta_{15} \Delta \text{INT}_{i,t+1} + \beta_{16} \Delta \text{NA}_{i,t} \\ & + \beta_{17} \Delta \text{NA}_{i,t+1} + \beta_{18} \Delta \text{MV}_{i,t+1} + \text{YearFE} + \text{FirmFE} + \varepsilon_{i,t} \end{aligned} \quad (2)$$

where MV represents the firm market value, used as the dependent variable. It is estimated as the sum of the market value of equity and the book value of debt scaled by total assets. Ex_Cash and CEO_Char are defined in Sections 4.2.1 and 4.2.2, respectively. The interaction variable CEO_Char×Ex_Cash is our main variable of interest, which reflects the interaction effect of each CEO characteristic (CEO gender, CEO tenure, CEO age, CEO educational level) on the value of excess cash holdings. Other control variables include earnings before extra items (EAR), research and development expenses (RD), common

dividend paid (DIV), interest expenses (INT), net assets (i.e. total assets net of cash (NA)) and the future change in MV. All these variables, except NA, are scaled by total assets (TA) in the regression model. Their variants (i.e. leads, lags and differences) also serve as MV's determinants (Pinkowitz *et al.*, 2006; Frésard and Salva, 2010; Ilyas *et al.*, 2023). The details of the variables' measurements are provided in Appendix Table A1.

ΔX_t in equation (2) indicates a change in X from $t - 1$ to t , whereas ΔX_{t+1} represents the change from period t to $t + 1$.

We have estimated equation (2) using ordinary least squares (OLS) regression, including firm and year-fixed effects, to address the concern about omitted variables, which could create an endogeneity issue. Additionally, following Petersen (2009), the standard errors were adjusted for heteroskedasticity using two-way clustering at the firm and year levels. It has been documented by Thompson (2011) that using firm and year clustering simultaneously can significantly enhance the results of panel data analysis.

4.4 Summary statistics and correlation matrix

Table 2 provides descriptive statistics on all the main and control variables used in the analysis. The mean (median) of firm value proxied by MV is approximately 3.262 (1.787), which is in line with the existing studies (e.g. Ilyas *et al.*, 2022; Ilyas *et al.*, 2023). The mean value of CEO_Fem is 0.086, indicating that women occupy 8.6% of CEO positions in our sample. The variable CEO_Edu indicates that 37.4% of the CEOs examined hold postgraduate degrees. The average excess cash holding (ExCash) is 18.7%, which is higher than the average excess cash holdings reported by previous studies. For instance, Belkhir *et al.* (2014), Ilyas *et al.* (2022) find an average excess cash holding of 7.72% and 14.6% in France and Pakistan, respectively. This indicates that Malaysian firms keep a significant amount of cash. The standard deviation of 1.132 for the variable ExCash

Variables	Mean	Median	SD	Min	Max	VIF
<i>Dependent variable</i>						
MV	3.262	1.787	4.573	0.641	37.942	
<i>Independent variables</i>						
CEO_Fem	0.086	0.000	0.156	0.000	1.000	1.56
CEO_Tn	0.500	1.000	0.501	0.000	1.000	1.31
CEO_Age	0.500	0.000	0.505	0.000	1.000	1.12
CEO_Edu	0.374	0.000	0.284	0.000	1.000	1.43
ExCash	0.187	0.094	1.132	0.000	32.746	1.56
<i>Control variables</i>						
EAR	0.008	0.012	0.467	−3.234	0.764	2.15
Δ EAR	0.007	0.003	0.224	−1.168	1.842	1.73
Δ EAR _{t+1}	0.012	0.004	0.196	−1.082	1.626	1.45
RD	0.083	0.000	0.173	0.000	0.463	1.28
Δ RD	0.004	0.000	0.042	−0.261	0.325	1.39
Δ RD _{t+1}	0.005	0.000	0.038	−0.207	0.264	1.18
DIV	0.023	0.004	0.051	0.000	0.216	1.24
Δ DIV	0.003	0.000	0.034	−0.083	0.162	1.16
Δ DIV _{t+1}	0.002	0.000	0.027	−0.068	0.131	1.08
INT	0.018	0.009	0.031	0.000	0.225	1.33
Δ INT	0.001	0.000	0.010	−0.058	0.074	1.28
Δ INT _{t+1}	0.001	0.000	0.014	−0.046	0.056	1.20
Δ NA	0.051	0.064	0.248	−1.343	0.764	1.47
Δ NA _{t+1}	0.082	0.068	0.237	−0.736	1.238	1.35
Δ MV _{t+1}	0.278	0.086	1.874	−7.436	12.462	1.23
Source: Authors' own creation						

indicates that firms differ significantly in terms of excess cash holding. Table 2 also provides information on the variance inflation factor (VIF) based on the baseline specification (equation (2)). All VIFs are low, with the highest being 2.15, so multicollinearity is less of a concern.

Table 3 shows the correlation between the main variables of the study. It is necessary to examine the fact that the explanatory variables must not correlate with each other, i.e. multicollinearity should not exist. Moreover, it should be non-zero. It was found that all the correlation values were less than the threshold point of 0.8, implying that multicollinearity is not an issue.

5. Results and discussion

5.1 Baseline results

The results from the analysis of equation (2) are presented in Table 4. A quick glance reveals that the coefficient of ExCash is significantly positive in all columns, indicating that cash is valuable, consistent with the findings of prior studies (e.g. Dittmar and Mahrt-Smith, 2007; Frésard and Salva, 2010; Ilyas et al., 2023). In Column 1 of Table 4, where the results based on the benchmark model are presented, the coefficient of the interaction variable CEO_Fem×ExCash is significantly positive, indicating that excess cash holdings are more valuable in firms with female CEOs than those with male CEOs. The result supports the hypothesized relationship with the expectations of RBV theory. This finding suggests that a female CEO is a valuable resource that boosts the value of excess cash holdings. Women leaders exhibit different leadership styles and investment behaviors, affecting financial resource allocation and firm value. As Hillman et al. (2007) argued, including female CEOs within the company’s leadership structure enhances managerial effectiveness and organizational efficiency through their unique skills, competencies, knowledge and leadership styles. This, in turn, is anticipated to foster improved corporate governance practices, therefore reducing the chances of wasting excess cash flow. Our result supports prior studies, such as those by Amore et al. (2014), who find that the presence of female CEOs may result in more efficient managerial decision-making. Ward and Forker (2017) find that female CEOs enhance financial management. Ullah et al. (2020) claim that female CEOs significantly improve investment efficiency.

In Column 2 of Table 4, where the effect of CEO tenure is considered, the coefficient of the interaction variable CEO_Tn×ExCash is significantly negative, suggesting that going from the lowest tercile of CEO tenure to the highest tercile, the marginal impact of excess cash holdings on firm value is decreasing. As the CEO’s tenure lasts longer, the value of excess cash accumulated within the company itself gets worse. This result is inconsistent with the prediction of H2. Thus, contrary to longer tenured CEOs, shorter-tenured CEOs are a significant source of efficient financial resource allocation that improves firm value. This finding contradicts the expectation of the RBV theory that CEOs with extended tenure

Table 3 Pearson correlation matrix						
	MV	CEO_Fem	CEO_Tn	CEO_Age	CEO_Edu	ExCash
MV	1					
CEO_Fem	0.107*	1				
CEO_Tn	−0.084*	0.008	1			
CEO_Age	−0.027	0.004	0.067	1		
CEO_Edu	0.203***	0.093*	−0.038	−0.042	1	
ExCash	0.346***	0.119**	0.071	0.066	0.001	1

Notes: This table presents Pearson correlations of the main variables used in the regressions. ***, ** and * denote the statistical significance at the 1, 5 and 10% levels, respectively

Source: Authors' own creation

Table 4 OLS regression on CEO characteristics and firms' excess cash valuation

Dependent variable: $MV_{i,t}$ independent variables	(1) Female CEO	(2) CEO tenure	(3) CEO age	(4) CEO educational level
CEO_Fem	0.346** (2.37)			
CEO_Fem×ExCash	0.752*** (3.73)			
CEO_Tn		−0.197* (−1.68)		
CEO_Tn×ExCash		−0.346** (−2.06)		
CEO_Age			−0.067* (−1.71)	
CEO_Age×ExCash			−0.235* (−1.83)	
CEO_Edu				0.229*** (2.96)
CEO_Edu×ExCash				0.432*** (4.28)
ExCash	0.738*** (3.64)	0.692*** (3.42)	0.812*** (3.83)	0.751*** (3.72)
<i>Control variables</i>				
EAR	−0.327*** (−3.44)	−0.319*** (−3.30)	−0.322*** (−3.33)	−0.326*** (−3.39)
ΔEAR	0.093*** (2.97)	0.088*** (2.91)	0.091*** (2.94)	0.095*** (3.01)
ΔEAR _{t+1}	−0.113** (−2.13)	−0.107** (−2.02)	−0.109** (−2.05)	−0.116** (−2.16)
RD	1.304* (1.77)	1.297* (1.70)	1.295* (1.67)	1.306* (1.78)
ΔRD	0.373** (2.02)	0.366* (1.94)	0.364* (1.92)	0.377** (2.08)
ΔRD _{t+1}	0.193** (2.31)	0.193** (2.31)	0.193** (2.31)	0.193** (2.31)
DIV	8.626** (2.23)	8.346** (2.04)	8.338** (1.98)	8.876** (2.36)
ΔDIV	−2.078 (−1.18)	−1.831 (−1.04)	−1.778 (−0.94)	−2.113 (−1.34)
ΔDIV _{t+1}	4.557** (2.46)	4.373** (2.21)	4.403** (2.30)	4.467** (2.38)
INT	−4.338** (−2.17)	−4.338** (−2.17)	−4.338** (−2.17)	−4.338** (−2.17)
ΔINT	−0.714 (−0.98)	−0.682 (−0.79)	−0.697 (−0.92)	−0.731 (−1.08)
ΔINT _{t+1}	0.492 (1.51)	0.473 (1.39)	0.481 (1.45)	0.512 (1.58)
ΔNA	0.276** (2.37)	0.262** (2.24)	0.267** (2.29)	0.281** (2.41)
ΔNA _{t+1}	0.883*** (3.11)	0.870*** (2.93)	0.875*** (2.96)	0.911*** (3.18)
ΔMV _{t+1}	−0.543*** (−3.61)	−0.543*** (−3.61)	−0.543*** (−3.61)	−0.543*** (−3.61)
Constant	1.537*** (5.32)	1.382*** (4.83)	1.643*** (5.43)	1.522*** (5.19)
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Observations	1098	822	804	1098
Adjusted R^2	0.474	0.468	0.462	0.481

Notes: This table presents the results of the OLS regressions of the impact of CEO characteristics on the value of excess cash holdings. Year- and firm-fixed effects are controlled in all the regressions. ***, ** and * denote the statistical significance at the 1, 5 and 10% levels, respectively. The *t*-statistics are in parentheses

Source: Authors' own creation

accumulate experience, expertise and knowledge, leading to better financial resource allocation. According to [Lee and Wang \(2022\)](#), the CEO is likely to become more entrenched during his/her tenure in office. Therefore, long-tenured CEOs in firms with excess cash reserves can waste excess liquidity, lowering the marginal value of cash. [Han et al. \(2017\)](#) draw a similar conclusion. They claim that CEOs who have longer tenure periods retain much more cash for running companies stably, and this result can be interpreted to mean that stock investors may think those CEOs do not manage cash well. In contrast, short-tenured CEOs are more likely to allocate excess cash to investment opportunities to provide long-term investment incentives, which are expected to generate potential value for shareholders.

In relation to the association between CEO age and the value of excess cash holdings, we document a negative and significant relationship (Column 3 of [Table 4](#)). Specifically, moving from the lowest tercile of CEO age to the highest tercile, the marginal impact of excess cash holdings on firm value decreases significantly. This result is not consistent with the expectation of *H3*. This finding indicates that older CEOs are not a valuable resource that improves firms' competitive advantages and performance. This result aligns with the view that older CEOs are more conservative and less adaptable to business environment changes than younger CEOs, which might negatively affect financial resource allocation

and firm performance (Bertrand and Schoar, 2003). CEOs with shorter career horizons, such as older CEOs, are more concerned with short-term benefits and, therefore, are more likely to participate in self-seeking behavior (Lee *et al.*, 2018). On the contrary, younger CEOs with longer career paths tend to consider their long-term profits and invest more efficiently than their older counterparts, suggesting that younger CEOs appear to be managing excess cash well for stock investors (Xie, 2015; Han *et al.*, 2017).

In Column 4 of Table 4, where the effect of CEO education is included, the coefficient of the interaction variable CEO_Edu×ExCash is positive and significant, suggesting that CEO education improves the value of excess cash holdings. This finding supports the hypothesis relationship with the expectation of RBV theory, indicating that the educational level is a major source for a firm to improve financial resource allocation and the value of excess cash holdings. The result shows that educational level is a valuable resource for firms, as it meets the criteria of resources outlined in the RBV theory, which are rare, valuable and inimitable, as highlighted by Barney (1991). Education helps CEOs make effective choices that enhance firm performance (Naseem *et al.*, 2020). A CEO with the proper education, knowledge and talents could be a source of competitive advantage, leading to efficient financial resource allocation that improves firm value (Orazalin *et al.*, 2016; Gupta *et al.*, 2021). Our result supports the studies conducted by Ahmad *et al.* (2022) and Suherman *et al.* (2023)), who report that education level boosts firm performance.

The coefficients of our control variables are similar to those documented by Ilyas *et al.* (2023), who study the impact of foreign institutional investors on the value of excess cash holdings.

5.2 Robustness tests

We conducted additional tests to validate the robustness of our preliminary findings. First, we re-estimate equation (2) with alternative measures of the excess cash holdings. In the second specification, we use the dynamic two-step generalized method of moments (system GMM) estimator to examine the causal relationship between CEO characteristics and the value of excess cash holdings.

5.2.1 Results with alternative measures of excess cash holdings. To test the robustness of our main findings, we repeat our analysis using an alternative measure of excess cash holdings. Following Jilani *et al.* (2023), we use the 75% and 90% quantiles of the levels of cash holdings to estimate the level of excessive liquidity. Thus, the robustness tests focus only on firms that hold the most excessive amounts of cash reserves and therefore have a much greater portion of total cash resources that can be wasted (Dittmar and Mahrt-Smith, 2007).

The regression results are contained in Table 5. The results of these analyses are similar to those of the main tests reported in Table 4. The findings confirm that the interaction coefficients between excess cash holdings and different CEO characteristics (gender and educational level) are positive and significant. Interestingly, the positive effect of CEO characteristics (gender and educational level) on the value of excess cash holdings further increased, both in magnitude and significance level, when we replaced ExCash with the 75% and 90% quantiles, respectively.

These coefficients provide similar but even stronger indications that CEO characteristics (gender and educational level) increase the value of excess cash. Thus, these CEO characteristics are valuable resources that enhance the value of excess cash holdings.

Moreover, the influences of CEO tenure and CEO age on the value of excess cash holdings are also in line with our initial results. We find negative and significant coefficients associated with the interaction variables CEO_Tn×Cash and CEO_Age×Cash considering

Table 5 Results with 75% and 90% quantiles of the cash holdings

Dependent Variable: $MV_{i,t}$ Independent variables		Cash_75		Cash_90	
CEO_Fem	0.413*** (2.96)		0.434*** (3.12)		
CEO_Fem×Cash	0.837*** (4.07)		0.855*** (4.21)		
CEO_In				0.273* (1.88)	
CEO_In×Cash		−0.264* (−1.76)		0.513*** (2.97)	
CEO_Age		−0.492*** (−2.88)			−0.090** (−2.32)
CEO_Age×Cash					−0.291** (−2.27)
CEO_Edu					
CEO_Edu×Cash			0.307*** (3.36)		0.322*** (3.52)
Cash			0.561*** (4.52)		0.577*** (4.68)
Constant	0.584*** (3.19)	0.649*** (3.04)	0.602*** (3.28)	0.574*** (3.31)	0.681*** (3.63)
Control variables	1.273*** (4.81)	1.135*** (4.13)	1.344*** (4.94)	1.224*** (4.42)	1.362*** (5.26)
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Observations	1098	822	1098	822	1098
Adjusted R^2	0.536	0.525	0.542	0.543	0.561

Notes: This provides the regression estimates using samples with the top 75th (CASH_75) and 90th (CASH_90) percentiles of cash holdings. Year- and firm-fixed effects are controlled in all the regressions. ***, **, and * denote the statistical significance at the 1, 5 and 10% levels, respectively. The t-statistics are in parentheses

Source: Authors' own creation

both the 75% and 90% quantiles. These findings demonstrate that the value of excess cash holdings is decreasing in firms with long-tenured CEOs and in those with older CEOs.

5.2.2 Endogeneity: system generalized method of moments. We address the endogeneity concern in the relationship between CEO characteristics and the value of excess cash holdings by using the dynamic two-step system GMM. This method offers robustness against issues such as unobserved heterogeneity, reverse causality and dynamic endogeneity within the explanatory variables (Roodman, 2009). Table 6 reports the result of the System GMM panel estimation following Arellano and Bover (1995) and Blundell and Bond (1998). It can be observed that the interaction between CEO characteristics (gender and educational level) and excess cash holdings exhibits significant positive coefficients, indicating that these CEO characteristics improve the value of excess cash holdings. With regard to CEO tenure and CEO age, we also find negative and significant coefficients associated with the variables CEO_Tn×Cash and CEO_Age×Cash, suggesting that excess cash holdings are less valuable in firms with long-tenured CEOs and those with older CEOs. Additionally, the statistical insignificance of the AR2 and Hansen J-statistics indicates no autocorrelation and validates the instruments in our models. These results are consistent with the findings of the baseline regression in Table 4.

6. Conclusion

The CEO of a company plays a crucial role in shaping the company's strategic policies, finances and investment decisions. Recognizing this role, the literature on investment and financial management has long advocated for considering senior management attributes when assessing corporate investment decisions. However, research regarding the implications of CEO characteristics for excess cash valuation remains limited. Therefore, this study aims to investigate whether and how different CEO characteristics influence the use and ultimate valuation of excess cash holdings. Following existing studies, four CEO attributes (gender, tenure, age and educational level) with both excess cash holdings (the amount of cash holdings above a predicted optimal level) and market-based performance

Table 6 GMM analyses on CEO characteristics and firms' excess cash valuation

Variable	(1) Female CEO	(2) CEO tenure	(3) CEO age	(4) CEO educational level
CEO_Fem	0.391*** (3.21)			
CEO_Fem×ExCash	0.788*** (4.07)			
CEO_Tn		−0.238** (−2.22)		
CEO_Tn×ExCash		−0.452*** (−2.94)		
CEO_Age			−0.081** (−2.11)	
CEO_Age×ExCash			−0.192*** (−2.82)	
CEO_Edu				0.283*** (3.46)
CEO_Edu×ExCash				0.512*** (4.50)
ExCash	0.557*** (3.71)	0.629*** (3.51)	0.646*** (3.94)	0.566*** (3.82)
Constant	0.743*** (3.53)	0.681*** (3.27)	0.477*** (3.14)	0.761*** (3.67)
Control variables	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Observations	1,098	822	804	1,098
AR(1) (<i>p</i> -value)	0.93 (0.001)	0.84 (0.001)	0.61 (0.001)	0.97 (0.001)
AR(2) (<i>p</i> -value)	1.78 (0.209)	1.67 (0.186)	1.45 (0.152)	1.86 (0.216)
Sargan test (<i>p</i> -value)	1.44 (0.502)	1.31 (0.483)	1.26 (0.474)	1.49 (0.515)
Hansen-J test (<i>p</i> -value)	0.43 (0.613)	0.37 (0.568)	0.32 (0.541)	0.47 (0.622)

Notes: This table presents the results of endogeneity tests. We use a dynamic GMM estimator using robust standard errors clustered at the firm level. Year- and firm-fixed effects are controlled in all the regressions. ***, ** and * denote the statistical significance at the 1, 5 and 10% levels, respectively. The *t*- and *z*-statistics are in parentheses

Source: Authors' own creation

(market value) are used. The analyses are made using OLS regression, robust regression and GMM across 285 non-financial Malaysian firms over a period of 7 years from 2015 to 2021 and present the results from the perspective of RBV theory.

We document strong and robust empirical evidence that female CEOs and CEOs' educational levels demonstrate a statistically significant positive correlation with the value of excess cash holdings. This result is reliable with the expectation of RBV theory that female CEOs and CEOs' educational levels are valuable resources that help to improve cash management and thereby significantly contribute to firms' performance.

However, CEO tenure and CEO age negatively impact a firm's excess cash valuation. This result indicates that companies with long-tenured CEOs and those with older CEOs tend to experience implications that lead to inefficient cash management, resulting in reduced firm value when compared to short-tenured CEOs and younger CEOs, respectively.

The implications of this study extend across various sectors, starting with its practical relevance for practitioners in the corporate realm. The study emphasizes how crucial it is to take CEO qualities into account when choosing leaders and developing succession plans at the practical level of corporate governance processes. The findings suggest that firms could benefit from prioritizing the appointment of female CEOs and CEOs with higher educational levels within their top management teams. Such a strategy could potentially lead to improved valuations associated with excess cash holdings. By actively seeking out and promoting diverse leadership talent, firms may foster a more dynamic and effective approach to cash management, transcending traditional metrics such as tenure or age.

These results can also be used by policymakers to guide programs that attempt to improve educational attainment and gender diversity in business leadership. Advocates for laws that facilitate the appointment of women and highly educated people to executive roles can help create a corporate environment that is more inventive and inclusive. In the long run, these kinds of projects could improve the way surplus financial reserves are used and raise economic output in general.

The study provides investors with insightful information about the possible relationship between CEO traits and company performance, especially with regard to measures for managing surplus capital. Businesses headed by female CEOs or CEOs with more education may be thought to be in a better position to maximize excess cash holdings, which could have an impact on portfolio allocation techniques and investment choices. This emphasizes how crucial it is to take CEO traits into account in addition to conventional financial measures when doing investment analysis.

Although this study has important implications, it also has some limitations that pave the way for future research. First, the study focuses solely on 173 non-financial firms listed in Bursa Malaysia over a specific period (2015–2021). This sample size might not fully represent the diversity of firms in Malaysia or other countries. Therefore, the generalizability of the findings to other contexts might be limited. In terms of timeframe, the study covers a relatively short period (2015–2021). Economic conditions, corporate governance practices and CEO characteristics might vary over time. Thus, the findings may not be applicable to different time periods. Second, other factors could influence both CEO characteristics and excess cash holdings, leading to spurious correlations. In particular, given the dominance of family businesses, the executives and board directorships of such firms are occupied mainly by family members of the substantial owners' group, which is likely to provide them with the requisite comfort and power.

Despite the limitations of this study, future recommendations should address several key recommendations to overcome the limitations identified in the study. First of all, longer term research would provide a more thorough grasp of how the traits of CEOs affect the valuation of surplus cash holdings by accounting for changes in both the state of the economy and corporate governance over time. Furthermore, broadening the study's focus to include

companies from other cultural backgrounds in addition to Malaysia will allow for cross-cultural analysis, revealing contextual variables and global patterns influencing this link. To improve the analysis' thoroughness, researchers ought to include a larger range of CEO traits and firm-specific variables. The field would advance further if creative approaches were used to determine causal linkages and provide useful advice for businesses looking to maximize excess cash management. This would help with both academic research and business decision-making.

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Further reading

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Table A1 Variables' measures

<i>Variables</i>	<i>Measures</i>
<i>Independent variable</i>	
MV	Firm value: It is calculated as the sum of the market value of equity and total liabilities over total assets
<i>Dependent variables</i>	
ExCash	Excess cash holdings: It represents cash holdings above the predicted level, which is the positive residual value estimated through the optimal cash holdings regression using equation (1)
CEO_Fem	Female CEO: it equals 1 if the CEO is a woman and 0 if the CEO is a man
CEO_Tn	CEO tenure: it equals 1 for the highest CEO tenure tercile and 0 for the lowest CEO tenure tercile
CEO_Age	CEO age: it equals 1 for the highest CEO age tercile and 0 for the lowest CEO age tercile
CEO_Edu	CEO education level: it equals 1 for postgraduates (master's and doctorate) and 0 for undergraduates and others
<i>Control variables</i>	
EAR	The ratio of earnings before extraordinary items to NA
ΔEAR	EAR at year t minus EAR at year $t - 1$ over NA
ΔEAR_{t+1}	EAR at year $t + 1$ minus EAR at year t over NA
RD	Research and development (R&D) expenditures over NA
ΔRD	RD at year t minus RD at year $t - 1$ over NA
ΔRD_{t+1}	RD at year $t + 1$ minus RD at year t over NA
DIV	Common dividends over NA
ΔDIV	DIV at year t minus DIV at year $t - 1$ over NA
ΔDIV_{t+1}	DIV at year $t + 1$ minus DIV at year t over NA
INT	The ratio of interest expenses to NA
ΔINT	INT at year t minus INT at year $t - 1$ over NA
ΔINT_{t+1}	INT at year $t + 1$ minus INT at year t over NA
ΔNA	Net assets at year t minus Net assets at year $t - 1$ over NA
ΔNA_{t+1}	NA at year $t + 1$ minus NA at year t over NA
ΔMV_{t+1}	MV at year $t + 1$ minus MV at year t over NA
Source: Authors' own creation	

Corresponding author

Moncef Guizani can be contacted at: m.guizani@arabou.edu.sa

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